

1 Calcule las constantes K_p , K_v , K_a y el error en estado estacionario para:

1.1 Inciso A

$$\mathbf{G}(s) = \frac{1}{s^2(s+12)}; \mathbf{H}(s) = Ks$$

Solución:

Obteniendo $G(s)H(s)$:

$$G(s)H(s) = \frac{Ks}{s^2(s+12)}$$

$$G(s)H(s) = \frac{K}{s(s+12)}$$

Obteniendo K_p :

$$K_p = \lim_{s \rightarrow 0} G(s)H(s)$$

$$= \left| \frac{K}{s(s+12)} \right|_{s=0}$$

$$= \frac{K}{(0)((0)+12)} = \frac{K}{0} = \boxed{\infty}$$

Obteniendo K_v :

$$K_v = \lim_{s \rightarrow 0} sG(s)H(s)$$

$$= \left| \frac{K}{s+12} \right|_{s=0}$$

$$= \frac{K}{(0)+12} = \boxed{\frac{K}{12}}$$

Obteniendo K_a :

$$K_a = \lim_{s \rightarrow 0} s^2G(s)H(s)$$

$$= \left| \frac{Ks}{s+12} \right|_{s=0}$$

$$= \frac{K(0)}{(0)+12} = \frac{0}{12} = \boxed{0}$$

Calculando el error en estado estacionario:

$$e_{ss} = \frac{1}{1 + K_p} = \frac{1}{1 + \infty} = \boxed{0}$$

$$e_{ss} = \frac{1}{K_v} = \frac{1}{\frac{K}{12}} = \boxed{\frac{12}{K}}$$

$$e_{ss} = \frac{1}{K_a} = \frac{1}{0} = \boxed{\infty}$$

1.2 Inciso B

$$\mathbf{G}(s) = \frac{\mathbf{K}(1 + 0.5s)}{s(s + 1)(2s + 1)(s^2 + s + 1)}; \mathbf{H}(s) = 1$$

Solución:

Obteniendo $G(s)H(s)$:

$$G(s)H(s) = \frac{K(1 + 0.5s)}{s(s + 1)(2s + 1)(s^2 + s + 1)}$$

Obteniendo K_p :

$$\begin{aligned} K_p &= \lim_{s \rightarrow 0} G(s)H(s) \\ &= \left| \frac{K(1 + 0.5s)}{s(s + 1)(2s + 1)(s^2 + s + 1)} \right|_{s=0} \\ &= \frac{K(1 + 0.5(0))}{(0)((0) + 1)(2(0) + 1)((0)^2 + (0) + 1)} = \frac{K}{0} = \boxed{\infty} \end{aligned}$$

Obteniendo K_v :

$$\begin{aligned} K_v &= \lim_{s \rightarrow 0} sG(s)H(s) \\ &= \left| \frac{K(1 + 0.5s)}{(s + 1)(2s + 1)(s^2 + s + 1)} \right|_{s=0} \\ &= \frac{K(1 + 0.5(0))}{((0) + 1)(2((0) + 1)((0)^2 + ((0) + 1)))} = \frac{0}{3} = \boxed{0} \end{aligned}$$

Obteniendo K_a :

$$\begin{aligned} K_a &= \lim_{s \rightarrow 0} s^2G(s)H(s) \\ &= \left| \frac{K(1 + 0.5s)s}{(s + 1)(2s + 1)(s^2 + s + 1)} \right|_{s=0} \\ &= \frac{K(1 + 0.5(0))(0)}{((0) + 1)((0) + 1)((0)^2 + (0) + 1)} = \frac{0}{3} = \boxed{0} \end{aligned}$$

Calculando el error en estado estacionario:

$$e_{ss} = \frac{1}{1 + K_p} = \frac{1}{1 + \infty} = \boxed{0}$$

$$e_{ss} = \frac{1}{K_v} = \frac{1}{0} = \boxed{\infty}$$

$$e_{ss} = \frac{1}{K_a} = \frac{1}{0} = \boxed{\infty}$$

1.3 Inciso C

$$\mathbf{G(s)} = \frac{\mathbf{1}}{\mathbf{s^2(s + 12)}}; \mathbf{H(s)} = \frac{\mathbf{5(s + 1)}}{\mathbf{s + 5}}$$

Solución:

Obteniendo $G(s)H(s)$:

$$G(s)H(s) = \frac{5(s+1)}{(5+s)(s^2+s+12)}$$

Obteniendo K_p :

$$\begin{aligned} K_p &= \lim_{s \rightarrow 0} G(s)H(s) \\ &= \left| \frac{5(s+1)}{(5+s)(s^2+s+12)} \right|_{s=0} \\ &= \frac{5((0)+1)}{(5+(0))((0)^2+(0)+12)} = \boxed{\frac{1}{12}} \end{aligned}$$

Obteniendo K_v :

$$\begin{aligned} K_v &= \lim_{s \rightarrow 0} sG(s)H(s) \\ &= \left| \frac{5s(s+1)}{(5+s)(s^2+s+12)} \right|_{s=0} \\ &= \frac{5(0)((0)+1)}{(5+(0))((0)^2+(0)+12)} = \frac{0}{60} = \boxed{0} \end{aligned}$$

Obteniendo K_a :

$$\begin{aligned} K_a &= \lim_{s \rightarrow 0} s^2G(s)H(s) \\ &= \left| \frac{5s^2(s+1)}{(5+s)(s^2+s+12)} \right|_{s=0} \\ &= \frac{5(0)^2((0)+1)}{(5+(0))((0)^2+(0)+12)} = \frac{0}{60} = \boxed{0} \end{aligned}$$

Calculando el error en estado estacionario:

$$e_{ss} = \frac{1}{1 + K_p} = \frac{1}{1 + \frac{1}{12}} = \boxed{\frac{12}{13}}$$

$$e_{ss} = \frac{1}{K_v} = \frac{1}{0} = \boxed{\infty}$$

$$e_{ss} = \frac{1}{K_a} = \frac{1}{0} = \boxed{\infty}$$

1.4 Inciso D

$$\mathbf{G(s)} = \frac{\mathbf{K(2s + 1)(4s + 1)}}{\mathbf{s^2(s^2 + s + 1)}}; \mathbf{H(s)} = \mathbf{1}$$

Solución:

Obteniendo $G(s)H(s)$:

$$G(s)H(s) = \frac{K(2s + 1)(4s + 1)}{s^2(s^2 + s + 1)}$$

Obteniendo K_p :

$$\begin{aligned} K_p &= \lim_{s \rightarrow 0} G(s)H(s) \\ &= \left| \frac{K(2s + 1)(4s + 1)}{s^2(s^2 + s + 1)} \right|_{s=0} \\ &= \frac{K(2(0) + 1)(4(0) + 1)}{(0)^2((0)^2 + (0) + 1)} = \frac{K + 1}{0} = \boxed{\infty} \end{aligned}$$

Obteniendo K_v :

$$\begin{aligned} K_v &= \lim_{s \rightarrow 0} sG(s)H(s) \\ &= \left| \frac{K(2s + 1)(4s + 1)}{s(s^2 + s + 1)} \right|_{s=0} \\ &= \frac{K(2(0) + 1)(4(0) + 1)}{(0)((0)^2 + (0) + 1)} = \frac{K + 1}{0} = \boxed{\infty} \end{aligned}$$

Obteniendo K_a :

$$\begin{aligned} K_a &= \lim_{s \rightarrow 0} s^2G(s)H(s) \\ &= \left| \frac{K(2s + 1)(4s + 1)}{s^2 + s + 1} \right|_{s=0} \\ &= \frac{K(2(0) + 1)(4(0) + 1)}{(0)^2 + (0) + 1} = \frac{K + 1}{1} = \boxed{K + 1} \end{aligned}$$

Calculando el error en estado estacionario:

$$e_{ss} = \frac{1}{1 + K_p} = \frac{1}{1 + \infty} = \boxed{0}$$

$$e_{ss} = \frac{1}{K_v} = \frac{1}{\infty} = \boxed{0}$$

$$e_{ss} = \frac{1}{K_a} = \frac{1}{K + 1} = \boxed{\frac{1}{K + 1}}$$

1.5 Inciso E

$$\mathbf{G(s)} = \frac{\mathbf{1000}}{\mathbf{s(s + 10)(s + 100)}}; \mathbf{H(s)} = \mathbf{1}$$

Solución:

Obteniendo $G(s)H(s)$:

$$G(s)H(s) = \frac{1000}{s(s + 10)(s + 100)}$$

Obteniendo K_p :

$$\begin{aligned} K_p &= \lim_{s \rightarrow 0} G(s)H(s) \\ &= \left| \frac{1000}{s(s + 10)(s + 100)} \right|_{s=0} \\ &= \frac{1000}{(0)((0) + 10)((0) + 100)} = \frac{1000}{0} = \boxed{\infty} \end{aligned}$$

Obteniendo K_v :

$$\begin{aligned} K_v &= \lim_{s \rightarrow 0} sG(s)H(s) \\ &= \left| \frac{1000}{(s + 10)(s + 100)} \right|_{s=0} \\ &= \frac{1000}{((0) + 10)((0) + 100)} = \frac{1000}{1000} = \boxed{1} \end{aligned}$$

Obteniendo K_a :

$$\begin{aligned} K_a &= \lim_{s \rightarrow 0} s^2 G(s)H(s) \\ &= \left| \frac{1000s}{(s + 10)(s + 100)} \right|_{s=0} \\ &= \frac{1000(0)}{((0) + 10)((0) + 100)} = \frac{0}{0} = \boxed{0} \end{aligned}$$

Calculando el error en estado estacionario:

$$e_{ss} = \frac{1}{1 + K_p} = \frac{1}{1 + \infty} = \boxed{0}$$

$$e_{ss} = \frac{1}{K_v} = \frac{1}{1} = \boxed{1}$$

$$e_{ss} = \frac{1}{K_a} = \frac{1}{0} = \boxed{\infty}$$

2 ¿Qué valor de K_v se requiere para que $e_{ss} \leq 30\%$?

$$\mathbf{G(s)} = \frac{1}{s(s + 50)}; \mathbf{H(s)} = \mathbf{K}$$

Solución:

Obteniendo $G(s)H(s)$:

$$\begin{aligned} G(s)H(s) &= \frac{1}{s(s + 10)} \cdot K \\ &= \boxed{\frac{K}{s(s + 10)}} \end{aligned}$$

A partir de e_{ss} :

$$\begin{aligned} e_{ss} &= \frac{1}{K_v} \Rightarrow K_v = \frac{1}{e_{ss}} \\ K_v &= \frac{1}{0.3\bar{3}} = 3.0\bar{3} \end{aligned}$$

3 Determine si los siguientes sistemas son estables:

3.1 Inciso A

$$s^7 + 3s^6 + 3s^5 + s^4 + s^3 + 3s^2 + 3s + 10$$

Solucion:

$$\begin{array}{c|cccc} s^7 & 1 & 3 & 1 & 3 \\ s^6 & 3 & 1 & 1 & 10 \\ s^5 & & & & \\ s^4 & & & & \\ s^3 & & & & \\ s^2 & & & & \\ s^1 & & & & \\ s^0 & & & & \end{array}$$

3.2 Inciso B

$$s^4 + s^3 + s^2 + s + 3$$

3.3 Inciso C

$$2s^4 + s^3 + 3s^2 + 5s + 10$$

4 Determine el valor de K para que los siguientes sistemas sean estables:

4.1 Inciso A

$$s^4 + 3s^2 + Ks^2 + 2s + 1 = 0$$

4.2 Inciso B

$$s^3 + 3408.3s^2 + 1204000s + 1.5 \times 10^{-4}K = 0$$

4.3 Inciso C

$$s^3 + 3Ks^2 + (K + 2)s + 4 = 0$$